Demand Sentinel: Global Gradient-Boosted Forecasting for Intermittent Retail Demand

Tweedie-objective cross-learning with direct quantile intervals, evaluated on 30,490 M5 store–item series under rolling-origin validation

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Source repository: https://github.com/rudraakshreddy/demand-sentinel
Deployed application: https://demand-sentinel.streamlit.app

Executive Summary

Replenishment decisions in grocery retail are made per product, per store, over a few weeks, for tens of thousands of pairs at once. At that granularity demand is *intermittent*: across the 30,490 M5 store–item series studied here, 68.0% of all series–day observations are exactly zero and mean daily sales are 1.13 units. Applying the Syntetos–Boylan taxonomy, 72.6% of series are intermittent and 18.2% lumpy, while only 6.3% are smooth — the regime for which conventional exponential smoothing and ARIMA models were designed. Demand Sentinel is built and evaluated for the other 93.7%.

Two decisions follow from that characterisation and define the approach. First, the training objective is Tweedie regression with variance power $p = 1.1$, which for $1 < p < 2$ is a compound Poisson–gamma distribution: a point mass at exactly zero plus a continuous positive density. This is the structure of the response, whereas a squared-error objective assumes a symmetric error about a possibly negative mean and requires negative forecasts to be clipped after the fact. Second, accuracy is measured by the root mean squared scaled error and its dollar-weighted aggregate, not by percentage error — MAPE is undefined for 68% of this dataset, and excluding zero actuals does not repair the measure but silently restricts the evaluation to the faster-moving third of the data.

A single global gradient-boosted model is trained across all 30,490 series jointly, on the argument that individual series are too short and too sparse to identify their own dynamics while calendar, price and promotional effects are shared and become estimable when pooled. Prediction intervals come from direct quantile regression at $\alpha = 0.025$ and 0.975 rather than from a symmetric band around the point forecast. Validation is rolling-origin with three non-overlapping 28-day horizons; every feature, scaling factor and aggregation weight is recomputed from training data only at each origin.

The global model is best on every measure, attaining RMSSE 0.718 and WRMSSE 0.797 against 0.761 / 0.871 for Croston–SBA — the reference method for intermittent demand — and 0.977 / 1.116 for the seasonal naive benchmark. Fold-to-fold spread in RMSSE is 0.033, and the advantage holds within every band of series sparsity rather than being concentrated in the dense series. The complete three-fold evaluation runs in 35.8 minutes on twelve CPU cores.

- Matching the objective to the response distribution, rather than correcting a generic objective afterwards, is the single most consequential modelling decision.
- Cross-learning across series beats methods purpose-built for intermittency: one global model outperforms Croston–SBA by 8.4% on the dollar-weighted metric, at lower operational cost than maintaining 30,490 local models.
- Metric choice is substantive, not administrative: MAPE, RMSSE and WRMSSE rank the two naive benchmarks in three different ways on identical forecasts.
- Prediction intervals over-cover, at 98.96% empirical against a nominal 95%, with the excess rising in series sparsity. This is reported as a calibration shortfall — wider intervals mean excess safety stock — and quantile recalibration is identified as the correction.

Keywords: demand forecasting; intermittent demand; global models; gradient boosting; Tweedie regression; quantile regression; RMSSE; M5 competition.

Contents

1 Introduction

1.1 Retail demand forecasting as an operational problem

Replenishment decisions in grocery retail are made at the level of an individual product in an individual store, over a horizon of a few weeks, for tens of thousands of such pairs simultaneously. The forecast is not an end in itself: it is an input to an inventory policy whose costs are asymmetric — a stockout forfeits margin and customer goodwill, while overstock consumes working capital and, for perishables, results in write-off. Two properties of this setting shape everything that follows.

The first is *intermittency*. At store–item granularity most products do not sell every day. In the data studied here 68.0% of all series–day observations are exactly zero, and the mean daily sales rate is 1.13 units. A demand process that is zero most of the time and occasionally positive is not adequately described by the Gaussian-error models that dominate introductory forecasting practice, and, more consequentially, it breaks the accuracy measures most commonly reported.

The second is *scale with structure*. The 30,490 series are not independent: they are nested within stores, states, departments and categories, they share calendar effects and promotional mechanisms, and individually each is too short and too noisy to support a well-identified univariate model. This is the configuration in which a single *global* model estimated across all series generally outperforms a library of per-series local models [8, 15].

1.2 Contributions

(i) A characterisation of the demand population using the Syntetos–Boylan taxonomy [21], establishing quantitatively which forecasting regime the majority of series occupy and therefore which methods and metrics are admissible (Section 3).

(ii) A global gradient-boosted model with a Tweedie objective, chosen because the compound Poisson–gamma family is the natural likelihood for a zero-inflated non-negative response, evaluated against four statistical baselines including Croston–SBA, the standard method for intermittent demand (Section 4).

(iii) Distributional forecasts via direct quantile regression [10] rather than Gaussian intervals, with calibration assessed by empirical coverage and sharpness by mean interval width and pinball loss (Sections 4 and 5).

(iv) An explicit demonstration that percentage-error metrics are inadmissible on this data, and that the scaled metric adopted by the M5 competition ranks methods differently (Sections 4.5 and 5).

(v) Evaluation over all 30,490 series under rolling-origin walk-forward validation with three non-overlapping 28-day horizons, with no series subsetting (Section 4.6).

1.3 Scope and boundaries

Within scope. A single public dataset (M5 / Walmart) covering three US states, ten stores and 3,049 products at daily frequency. A 28-day forecast horizon, matching the M5 specification and typical replenishment cycles. Point forecasts and 95% prediction intervals at store–item granularity. Evaluation on scaled error metrics, dollar-weighted aggregate error, interval calibration and sharpness. A global tree-ensemble model and four statistical baselines.

Outside scope. *Hierarchical reconciliation:* forecasts are produced and evaluated at the bottom level only; coherence with store, department, state and total aggregates is not enforced. *Deep sequence models:* recurrent and attention architectures such as DeepAR [17] and its successors are not evaluated. *Inventory optimisation:* the mapping from a forecast distribution to an order quantity, safety stock, or service level is not modelled; the study stops at the forecast. *Price and promotion optimisation:* prices are treated as exogenous observed covariates, not as decision variables, so no elasticity or causal claim is made. *Cross-sectional cannibalisation and substitution:* interactions between competing products are not modelled beyond what the shared global model implicitly absorbs. *New-product cold start:* series are required to have an observed sales history; launch forecasting is not addressed. *Other retail settings:* generalisation to non-grocery, online, or non-US retail is untested.

2 Related Work

2.1 Intermittent demand

The foundational method is that of Croston [3], which decomposes an intermittent series into separate exponentially smoothed estimates of demand size and inter-arrival interval, forecasting their ratio. Syntetos and Boylan [20] showed Croston's estimator to be biased and proposed the Syntetos–Boylan approximation applying a factor of $1 - \alpha/2$, which is the variant used here. Syntetos et al. [21] introduced the taxonomy — smooth, erratic, intermittent, lumpy — that partitions series by average inter-demand interval and the squared coefficient of variation of non-zero demand; this classification is applied in Section 3 to establish which regime dominates. Subsequent work has examined state-space formulations [19], neural approaches [12], and forecast combination [16]. Kolassa [11] argues specifically that count-data distributions, not point forecasts, are the appropriate object in retail.

2.2 Global models and competition evidence

The M4 [13] and M5 [14] competitions provide the strongest available evidence on comparative accuracy. The M5 result relevant here is that gradient-boosted decision trees trained globally across all series dominated both statistical benchmarks and most neural approaches, and that the top solutions relied on cross-learning rather than per-series estimation. Montero-Manso and Hyndman [15] give a theoretical account: global models trade a small increase in bias for a large reduction in variance, which is favourable exactly when individual series are short and noisy. Januschowski et al. [8] provide the taxonomy of local versus global and of objective-driven versus data-driven methods within which the present model is classified.

2.3 Accuracy measurement

Hyndman and Koehler [7] established that percentage errors are undefined at zero actuals and asymmetric elsewhere, and proposed scaled errors as the general-purpose alternative. The M5 competition adopted the root mean squared scaled error, weighted by cumulative dollar sales, as its official metric [14]. For probabilistic forecasts, Gneiting and Raftery [5] formalise proper scoring rules and Gneiting et al. [6] the principle of maximising sharpness subject to calibration — the criterion applied to the intervals in Section 5. On validation design, Tashman [22] and Bergmeir and Benítez [1] establish that rolling-origin evaluation is required for temporally ordered data.

3 Data

3.1 Structure

The M5 dataset comprises unit sales of 3,049 products across ten Walmart stores in California, Texas and Wisconsin, observed daily for 1,941 days from 29 January 2011 to 19 June 2016. Products are organised into three categories (Hobbies, Household, Foods) and seven departments. The store–item cross-product gives 30,490 individual series, all of which are used; no subsetting or filtering is applied.

Three tables are combined. The sales table supplies the response; a calendar table supplies dates, weekday indices, month, year, two channels of named events (religious, national, cultural and sporting), and SNAP benefit-eligibility flags that differ by state; a price table supplies weekly per-store item prices, from which promotional activity is inferred.

3.2 Intermittency and demand classification

Across all 59.2M series–day observations, 68.0% are zero and the mean is 1.13 units per day. The distribution of per-series zero-fraction is broad rather than concentrated, so the population contains both

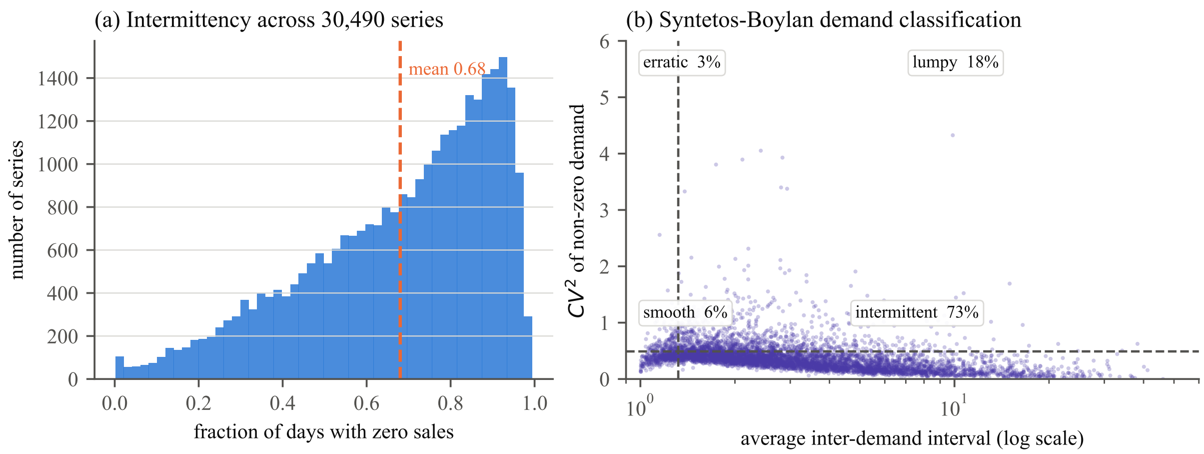

Figure 1. Demand structure across all 30,490 series. (a) Distribution of the per-series fraction of days with zero sales. (b) Syntetos–Boylan classification on average inter-demand interval and squared coefficient of variation of non-zero demand; dashed lines are the conventional cutoffs at 1.32 and 0.49. Percentages give the share of series in each quadrant.

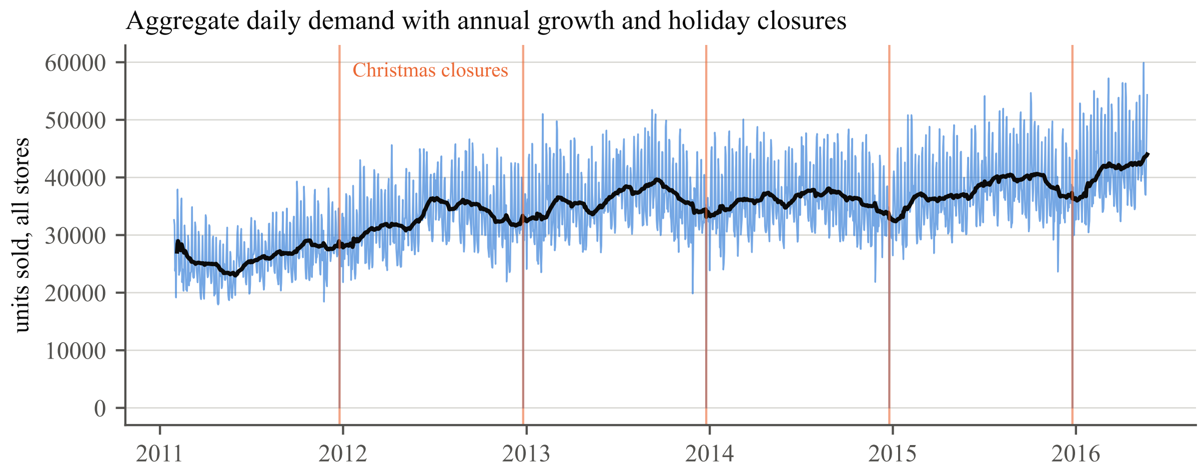

Figure 2. Aggregate daily unit sales across all stores, with a 28-day moving average. Growth over the period is steady and weekly seasonality is strong; the sharp annual troughs are Christmas Day, the one day on which the stores close.

fast-moving and near-dormant products and cannot be treated homogeneously.

Figure 1 applies the Syntetos et al. [21] classification, which partitions series on the average inter-demand interval (ADI) at a cutoff of 1.32 and the squared coefficient of variation of non-zero demand sizes (CV^2) at 0.49. The relevant consequence is that only a minority of series fall in the *smooth* quadrant for which conventional exponential smoothing and ARIMA models were designed; the majority are erratic, intermittent or lumpy. Method and metric selection must be made for that majority, not for the convenient minority.

At the aggregate level (Figure 2) the series is well behaved: a clear upward trend, pronounced weekly seasonality, and sharp, fully predictable annual troughs at Christmas closures. The contrast between this and the bottom-level intermittency is the central modelling tension. Signal that is obvious in aggregate is buried in noise at the granularity at which decisions are actually made, which is the argument for a global model that can borrow the aggregate structure while predicting at the bottom level.

4 Methodology

4.1 Feature construction

The panel is reshaped to long form, one row per series–day, and features are constructed within each series in strict chronological order. Every feature is a function of information available strictly before the target day; all lag and rolling operations are applied through grouped shifts so that no value from day t or later enters the row for day t.

Autoregressive lags. Sales at lags $\{1, 2, 3, 7, 14, 21, 28\}$. Short lags capture carry-over and promotional bursts; the weekly multiples capture the strong day-of-week cycle visible in Figure 2. The longest lag equals the forecast horizon, which is the minimum lag guaranteed to be observable at prediction time across the whole horizon.

Rolling statistics. Means and standard deviations over $\{7, 14, 28\}$-day windows, computed on the lagged series rather than the contemporaneous one. The means summarise the local level; the standard deviations supply a local dispersion estimate, which is informative precisely because dispersion varies so strongly across the intermittency spectrum.

Exponentially weighted means with $\alpha \in \{0.3, 0.1\}$, giving two rates of recency emphasis and a smoother level estimate than a rectangular window on sparse data.

Calendar. Day of week, month, day of year, an event indicator, the state-specific SNAP flag, and the interaction of SNAP with day of week. The interaction is included because benefit disbursement follows a fixed within-month schedule that differs by state and interacts with normal weekly shopping patterns.

Price. The current shelf price, its week-on-week change, and its ratio to a trailing 28-day mean. A promotion flag is set when that ratio falls below 0.95. Price is used as an observed exogenous covariate; no causal or elasticity interpretation is claimed.

Identity. Item, department, category, store and state, encoded as integer codes shared globally across all shards so that the same physical entity receives the same code everywhere. These allow the global model to learn series-specific and group-specific levels without fitting separate models.

Rows in the first 28 days of each series, for which the longest lag is undefined, are discarded.

4.2 Global gradient-boosted model

A single gradient-boosted tree ensemble [2] is trained across all series jointly. The objective is Tweedie regression with variance power $p = 1.1$,

$$\mathrm{Var}(Y) = \phi\, \mu^{p}, \qquad 1 < p < 2, \tag{1}$$

which for p in this range corresponds to a compound Poisson–gamma distribution: a positive probability mass at exactly zero, together with a continuous positive density [9, 18]. This is the natural likelihood for the response at hand, and it is the principal modelling decision in this study. A squared-error objective would implicitly assume a symmetric Gaussian error around a possibly negative mean, which is both distributionally wrong for a count that is zero 68% of the time and operationally wrong, since it produces negative forecasts that must then be clipped.

Training uses a trailing 730-day window — two complete annual cycles — across all 30,490 series, giving approximately 22.3M training rows. Trees are grown to depth 8 with a learning rate of 0.06, row and column subsampling at 0.8, a minimum child weight of 20, and 250 boosting rounds. The minimum child weight is set relatively high deliberately: on sparse data, small leaves would otherwise fit individual zero runs.

4.3 Prediction intervals by quantile regression

Prediction intervals are produced by two additional models trained with the pinball objective [10] at $\alpha = 0.025$ and $\alpha = 0.975$,

$$L_\alpha(y, \hat{q}) \;=\; \max\bigl(\alpha(y - \hat{q}),\; (\alpha - 1)(y - \hat{q})\bigr), \tag{2}$$

whose minimiser is the conditional α-quantile. Estimating the bounds directly is preferred over adding a multiple of a residual standard deviation to the point forecast, because the latter presumes a symmetric error distribution — an assumption that fails badly for a response with a large point mass at zero. Direct quantile estimation allows the interval to be asymmetric and, where the conditional distribution warrants, to have a lower bound of exactly zero.

4.4 Statistical baselines

Four baselines are computed directly from the raw sales matrix, so their values are exact rather than approximations.

Naive repeats the last observed value across the horizon. *Seasonal naive* repeats the last seven observed values, tiling them across the 28-day horizon; given the strength of weekly seasonality this is the demanding benchmark and is the reference against which the scaled metric is interpreted. *Simple exponential smoothing* with $\alpha = 0.2$ supplies a smoothed level forecast. *Croston–SBA* [3, 20] maintains separately smoothed estimates $\hat{z}$ of non-zero demand size and $\hat{p}$ of the inter-demand interval, both with $\alpha = 0.1$, and forecasts $(1 - \alpha/2)\, \hat{z}/\hat{p}$; it is included because it is the reference method for precisely the demand regime that dominates this dataset.

4.5 Accuracy measures

Scaled error. The primary measure is the root mean squared scaled error, in which the forecast error is normalised by the in-sample one-step error of the naive method on the same series,

$$\mathrm{RMSSE}_i = \sqrt{\frac{\frac{1}{h}\sum_{t=n+1}^{n+h}\left(y_{i,t} - \hat{y}_{i,t}\right)^2}{\frac{1}{n-1}\sum_{t=2}^{n}\left(y_{i,t} - y_{i,t-1}\right)^2}}. \tag{3}$$

The denominator is a root-mean-square of first differences, not a mean absolute difference; matching the order of the numerator is what makes the ratio dimensionless and scale-free. Following the M5 convention the denominator is accumulated only from each series' first non-zero observation, so products not yet launched do not contribute a degenerate scale. A value below one indicates performance better than the in-sample naive benchmark.

Weighted aggregate. Series are aggregated by cumulative dollar sales over the final 28 training days,

$$\mathrm{WRMSSE} = \sum_{i=1}^{N} w_i\, \mathrm{RMSSE}_i, \qquad w_i = \frac{\sum_{t\in\text{last }28} y_{i,t}\, p_{i,t}}{\sum_j \sum_{t\in\text{last }28} y_{j,t}\, p_{j,t}}. \tag{4}$$

This reflects the operational reality that an error on a high-revenue product matters more than the same proportional error on a slow-moving one. Both the unweighted mean RMSSE and the weighted WRMSSE are reported, since they answer different questions.

Why percentage errors are excluded. The mean absolute percentage error is undefined whenever the actual is zero, which is the case for 68% of observations here. The conventional remedy — excluding zero actuals — does not repair the measure, it changes the estimand: the resulting statistic describes performance on the non-zero subpopulation, which is systematically composed of the faster-moving series. MAPE is nevertheless computed and reported in Section 5 on exactly that restricted basis, in order to demonstrate that it induces a different ranking of methods than the scaled metric does.

Probabilistic measures. Interval quality is assessed by empirical coverage, the fraction of actuals falling inside the nominal 95% interval; by mean interval width as a measure of sharpness; and by the pinball loss of Equation (2) evaluated at each bound, which is a proper scoring rule for the corresponding quantile [5].

4.6 Validation protocol

Evaluation uses rolling-origin walk-forward validation [1, 22] with three non-overlapping test windows of 28 days each, at training origins ending on days 1,857, 1,885 and 1,913. For each fold the model is retrained

Table 1. Forecast accuracy averaged over three rolling-origin folds, all 30,490 series, 28-day horizon. Lower is better throughout. RMSSE and WRMSSE are scale-free; a value below one indicates performance better than the in-sample naive benchmark.

Model	RMSSE	WRMSSE	MAE	RMSE	MAPE†
XGBoost (global, Tweedie)	**0.718**	**0.797**	**0.967**	**1.941**	**0.557**
Croston–SBA	0.761	0.871	1.082	2.282	0.587
Simple exp. smoothing	0.768	0.893	1.071	2.299	0.628
Seasonal naive (7)	0.977	1.116	1.229	2.732	0.837
Naive	0.976	1.184	1.372	3.111	0.916

Note. †MAPE is computed only on the 32% of observations with a non-zero actual and is reported solely for the comparison in Section 5.3; it is not a valid summary of performance on this dataset.

from scratch on data up to the origin, and every feature, scaling factor and aggregation weight is recomputed from training data only. Random k-fold cross-validation is not used: it would place future observations in the training set, and on a panel with shared calendar effects it would also leak information across series.

5 Results

5.1 Accuracy

Table 1 reports accuracy averaged across the three folds. The global model is best on every measure. Against Croston–SBA — the reference method for intermittent demand and the strongest statistical competitor here — it reduces RMSSE by 5.6% and WRMSSE by 8.4%. Against the seasonal naive benchmark it reduces RMSSE by 26.5% and WRMSSE by 28.5%.

Two features of the table deserve comment.

First, the ordering of the four statistical baselines is itself informative. Croston–SBA and simple exponential smoothing both clearly outperform the naive methods, which is the expected result when 90.8% of series are intermittent or lumpy (Section 3): methods that estimate a smoothed *rate* rather than reproducing recent observations are better matched to sparse data. That Croston–SBA does not dominate SES on the unweighted metric, despite being purpose-built for intermittency, is consistent with the mixed findings in the intermittent-demand literature [16].

Second, the gap between RMSSE and WRMSSE is not constant across models. For the global model the two are 0.718 and 0.797; for the naive method they are 0.976 and 1.184. Because WRMSSE weights by dollar sales, a larger WRMSSE-to-RMSSE ratio indicates that a method performs relatively worse on high-revenue series. The naive method degrades most under weighting, and the global model degrades least — an operationally relevant distinction that the unweighted metric alone would conceal.

Fold-to-fold variation is small. Across the three folds the global model attains RMSSE of 0.704, 0.712 and 0.737, a spread of 0.033, so the ranking in Table 1 is not an artefact of a particular test window.

5.2 Where the accuracy comes from

Figure 4 decomposes error by sparsity. Two things are visible. Accuracy degrades monotonically with intermittency for both methods, which is expected: a series that sells on one day in ten carries less information per unit of history. More importantly, the ranking is preserved in every band. The global model's advantage is not concentrated in the dense, well-behaved series where any method performs adequately; it holds in the sparse bands, which constitute the majority of the population and the bulk of the operational difficulty.

Figure 5 attributes the model's splits. The concentration on recent lags and rolling means confirms that the dominant signal is local level and recent momentum, with weekly lags supplying the day-of-week structure. Price features and the SNAP interaction contribute materially, which validates their inclusion; the identity

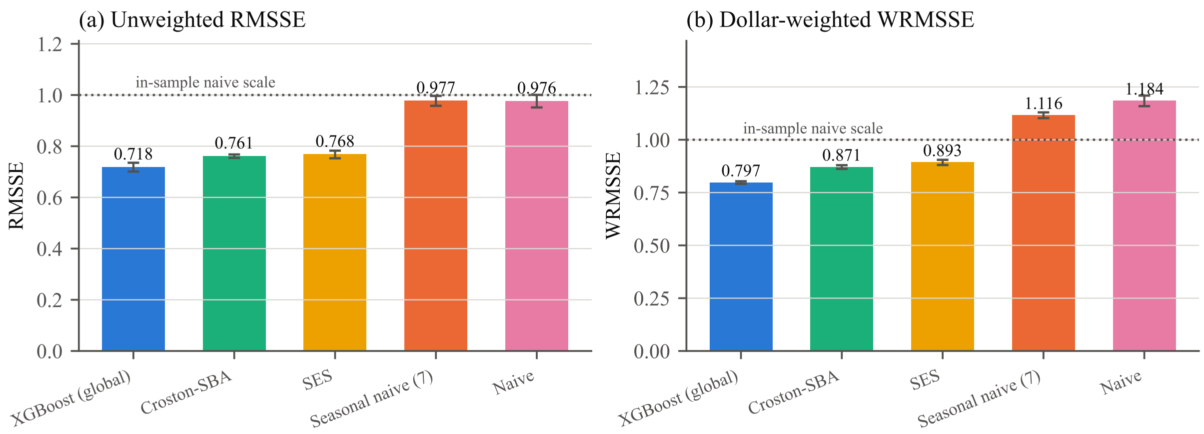

Figure 3. Forecast accuracy by model, mean over three folds with error bars giving the fold standard deviation. (a) Unweighted RMSSE. (b) Dollar-weighted WRMSSE. The dotted line at unity is the in-sample naive scale: values below it indicate a method that beats the naive benchmark on its own series.

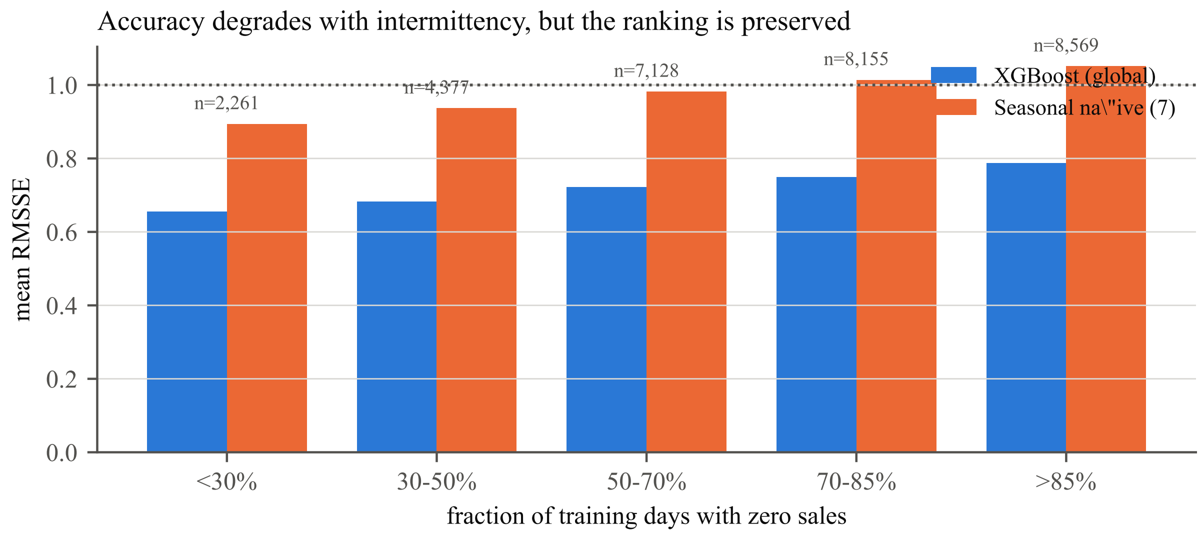

Figure 4. Mean RMSSE within bands of series sparsity, third fold. Accuracy degrades monotonically as the proportion of zero-sales days rises, for both methods, but the ordering is preserved throughout: the advantage of the global model is not confined to the easy, fast-moving series.

features allow the single global model to carry series-specific and group-specific levels without per-series estimation, which is precisely the mechanism by which cross-learning substitutes for the short individual histories.

5.3 Percentage error induces a different ranking

The concrete consequence of using a percentage error on this data is visible in the two naive methods. MAPE separates them clearly, at 0.837 for the seasonal variant against 0.916 for the plain one. RMSSE places them level, at 0.977 and 0.976. WRMSSE reverses the ranking, at 1.116 and 1.184 — so on the dollar-weighted measure the seasonal method is better, on the unweighted scaled measure they are indistinguishable, and on MAPE the gap is large.

The reason is structural rather than incidental. MAPE is evaluated on the 32% of observations with a non-zero actual, and that subpopulation is not a random sample: it over-represents fast-moving series and excludes exactly the sparse behaviour that distinguishes the methods. A model selected on MAPE would be selected on its performance over a third of the data, chosen by a criterion correlated with the outcome. This

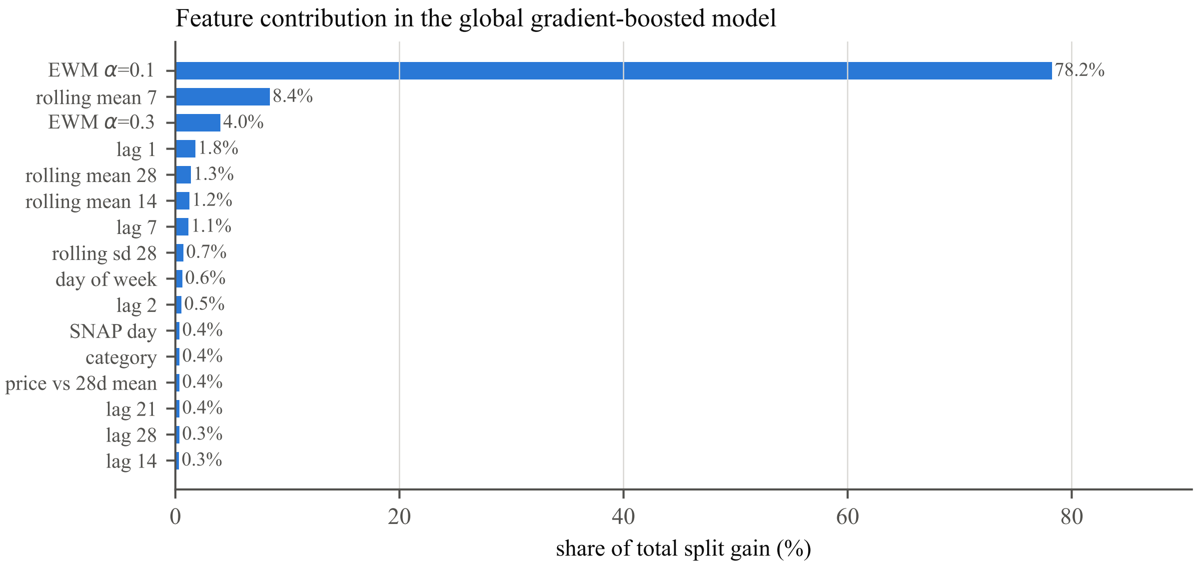

Figure 5. Feature contribution by share of total split gain in the global model. Recent lags and rolling means dominate, followed by price and identity features.

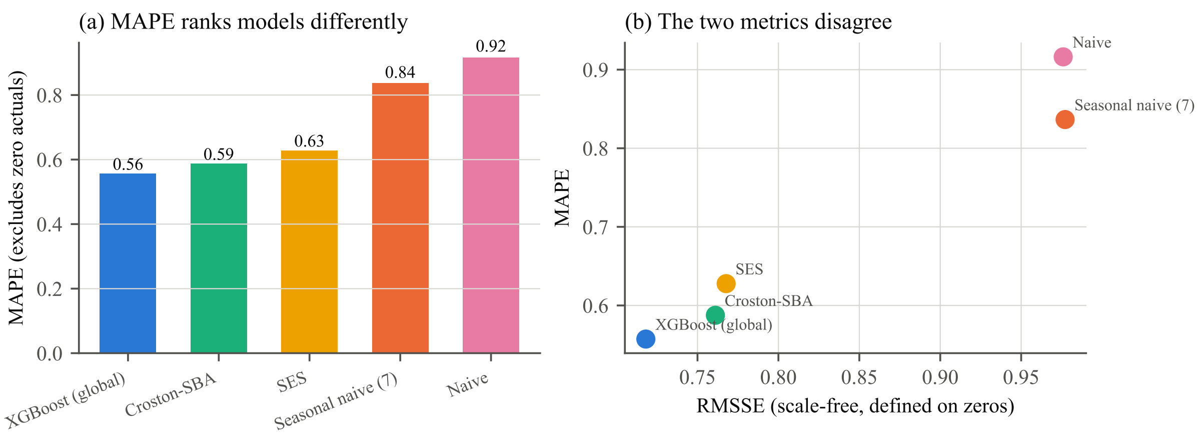

Figure 6. (a) MAPE by model, computed only on observations with a non-zero actual. (b) MAPE against RMSSE. The two measures do not agree: MAPE ranks the seasonal naive method ahead of the naive method, while RMSSE places them level and WRMSSE reverses the order.

is why the M5 competition adopted a scaled metric [14], and why RMSSE and WRMSSE are treated as primary here.

5.4 Prediction intervals

The intervals are *over-covering*: empirical coverage is 98.96% against a nominal 95%. Figure 7 shows the effect is systematic and increases with sparsity, from 96.1% coverage in the densest band to 99.8% in the sparsest.

This is reported as a calibration shortfall rather than presented as a strength. Following Gneiting et al. [6], the objective is to maximise sharpness *subject to* calibration; an interval that covers 99% of outcomes when 95% was requested is not better-calibrated, it is wider than necessary, and for inventory purposes an over-wide upper bound translates directly into excess safety stock. The mechanism is identifiable: for a series that is zero on most days, the conditional 97.5th percentile is estimated from very few informative observations, and the pinball objective under sparse positive support resolves toward a conservative bound. The narrower widths in the sparsest bands (4.74 units versus 12.42 in the densest) confirm that the model

Table 2. Prediction-interval quality for the global quantile models, nominal 95% coverage, averaged over three folds.

Fold	Empirical coverage	Mean width (units)	Pinball loss (lower / upper)
1	98.95%	6.27	0.0338 / 0.1544
2	99.00%	6.26	0.0338 / 0.1509
3	98.93%	6.29	0.0353 / 0.1529
Mean	98.96%	6.27	0.0343 / 0.1527

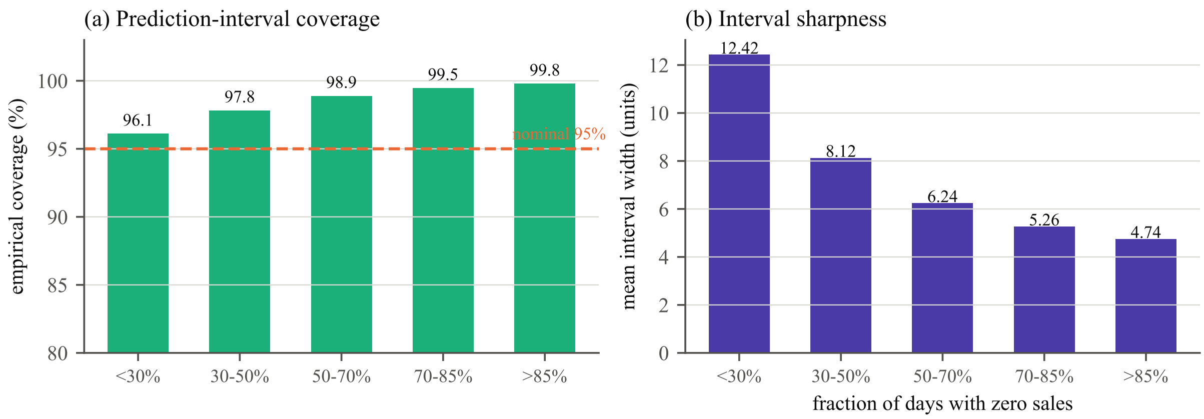

Figure 7. Interval behaviour by series sparsity, third fold. (a) Empirical coverage against the nominal 95% level. (b) Mean interval width. Coverage exceeds nominal in every band and rises with sparsity, while width falls — the intervals are conservative, and increasingly so for the sparsest series.

does adapt its interval to the series; it simply does not tighten fast enough to hit the nominal level.

A practical correction — not applied here, since it would require a further held-out split — would be to recalibrate the quantile levels on a validation fold, fitting the nominal levels that deliver 95% empirical coverage rather than assuming that requesting 0.025 and 0.975 delivers it.

5.5 Computational cost

The complete evaluation — three folds, each retraining a point model and two quantile models on approximately 8.9M rows drawn from all 30,490 series — completes in 35.8 minutes on twelve CPU cores with no GPU. Per fold, feature assembly takes roughly one minute, the point model trains in 126–207 seconds, and each quantile model in 118–158 seconds. The four statistical baselines, computed exactly over the full history of every series, take 115–217 seconds per fold.

This matters for the practical claim. A single global model covering 30,490 series retrains in minutes, whereas fitting and maintaining 30,490 individual univariate models — each requiring order selection and diagnostic checking — is not operationally comparable. The accuracy advantage in Table 1 is therefore obtained at lower, not higher, operational cost.

6 Discussion

6.1 Interpretation

Three findings carry beyond this dataset.

The first concerns the match between objective function and response distribution. The single most consequential modelling decision was the Tweedie objective, chosen because the compound Poisson–gamma family reproduces the structure of the response — a point mass at zero plus a positive continuous component. This is not a tuning choice; it is a statement about the data-generating process, and it removes the need for the post-hoc clipping of negative forecasts that a squared-error objective requires. The general principle is that where the response has known structural properties — non-negativity, a mass at zero, integer support — an objective respecting them should be preferred over a generic one with corrections applied afterwards.

The second concerns metric selection as a substantive rather than administrative decision. Section 5.3 shows two methods being ranked differently by MAPE, RMSSE and WRMSSE on the same forecasts. A metric that is undefined on 68% of observations does not become valid by discarding them; it silently changes the population under study. Choosing the accuracy measure is part of specifying the problem, and on intermittent data it determines the answer.

The third concerns global models. A single model trained across all series outperformed methods designed specifically for the dominant demand regime, including Croston–SBA. The mechanism is cross-learning: each individual series is too short and too sparse to identify its own dynamics, but the calendar, price and promotional effects are shared, and pooling makes them estimable [15]. The identity features then let the pooled model recover series-specific levels. This is the finding the M5 competition established at scale [14], and it reproduces here under an independent implementation.

6.2 Limitations

Single dataset. One retailer, three US states, ten stores, 2011–2016. No claim of generality to other retail formats, geographies or periods is made.

Training window and subsampling. The global model trains on a trailing 730-day window with rows subsampled at 40% for tractability within the available memory. All 30,490 series remain represented and the subsampling is uniform over rows, but the model does not see the complete history, and a fuller training set would likely improve accuracy somewhat.

Interval calibration. As reported above, intervals over-cover at 98.96% against a nominal 95%. The quantile levels were not recalibrated on a validation fold.

Fixed hyperparameters. Depth, learning rate and regularisation were set once and held across all folds rather than tuned per fold. Reported accuracy is therefore not the maximum attainable from this model class.

Bottom-level only. Forecasts are produced and evaluated at store–item granularity, with no reconciliation against store, department, state or total aggregates. A coherent hierarchy would be required for planning use.

Prices treated as known. Weekly prices over the forecast horizon are supplied to the model. This is realistic for planned promotions but optimistic where a retailer reprices reactively.

No statistical test of the accuracy differences. Differences in Table 1 are reported as point estimates over three folds. A Diebold–Mariano test [4] on the loss differentials would place the comparison on an inferential footing; three folds is too few to support one convincingly.

6.3 Future work

The immediate extensions follow the limitations. Quantile recalibration on a validation fold would correct the over-coverage documented in Table 2 at essentially no modelling cost. Hierarchical reconciliation would make the forecasts coherent across the aggregation structure and is a prerequisite for planning use. Training on the full history without row subsampling, together with per-fold hyperparameter search, would establish how much of the remaining gap is attributable to the training budget rather than the model class. A comparison against DeepAR [17] and modern transformer forecasters would test whether the global tree ensemble's advantage persists against global neural models rather than against local statistical ones. Finally, extending the pipeline from a forecast distribution to an inventory policy — newsvendor order quantities under the estimated quantiles, evaluated on realised service level and holding cost — would connect the forecast to the decision it exists to support, and would give the interval calibration issue a direct monetary interpretation.

6.4 Remarks

Demand Sentinel is presented as a reproducible benchmark study rather than a deployed planning system. Its accuracy figures are obtained on a public dataset under a protocol that is stated in full, and its two weakest points — interval over-coverage and the absence of hierarchical coherence — are documented above rather than omitted.

The broader observation concerns what the difficulty of this problem actually consists of. The aggregate demand series in Figure 2 is smooth, trending and strongly seasonal — an easy forecasting problem. The same demand at the granularity where replenishment decisions are made is zero 68% of the time. The forecasting difficulty is created by the disaggregation, not by the demand process, and the effective response is to pool information back across series while predicting at the level where it is needed.

7 Conclusion

Demand Sentinel evaluates retail demand forecasting across all 30,490 M5 store–item series under rolling-origin validation with three non-overlapping 28-day horizons. The demand population is characterised using the Syntetos–Boylan taxonomy: 72.6% of series are intermittent and 18.2% lumpy, with only 6.3% smooth, and 68.0% of all series–day observations are exactly zero.

A single global gradient-boosted model with a Tweedie objective — selected because the compound Poisson–gamma family matches a response with a point mass at zero — attains RMSSE 0.718 and WRMSSE 0.797, outperforming Croston–SBA (0.761 / 0.871), simple exponential smoothing (0.768 / 0.893) and the seasonal naive benchmark (0.977 / 1.116). The advantage is preserved within every band of series sparsity, so it does not derive from the dense series alone. Fold-to-fold variation is 0.033 in RMSSE, and the full three-fold evaluation completes in 35.8 minutes on twelve CPU cores.

Prediction intervals from direct quantile regression over-cover, at 98.96% empirical against a nominal 95%, with the excess increasing in series sparsity. This is a calibration shortfall, reported as such: the intervals are wider than requested, which for inventory purposes means excess safety stock, and recalibrating the quantile levels on a validation fold is the indicated correction.

Finally, the choice of accuracy metric is shown to be substantive rather than administrative. MAPE, which is undefined for 68% of the observations and is computed here on the remaining third, ranks the two naive benchmarks differently from RMSSE, and WRMSSE reverses that ordering again. On intermittent demand the metric selects the answer, and a scaled metric is the defensible choice.

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References

[1] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012.

[2] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[3] J. D. Croston. Forecasting and stock control for intermittent demands. *Operational Research Quarterly*, 23(3):289–303, 1972.

[4] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

[5] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[6] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B*, 69(2):243–268, 2007.

[7] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006.

[8] Tim Januschowski, Jan Gasthaus, Yuyang Wang, David Salinas, Valentin Flunkert, Michael Bohlke-Schneider, and Laurent Callot. Criteria for classifying forecasting methods. *International Journal of Forecasting*, 36(1):167–177, 2020.

[9] Bent Jorgensen. Exponential dispersion models. *Journal of the Royal Statistical Society: Series B*, 49 (2):127–162, 1987.

[10] Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

[11] Stephan Kolassa. Evaluating predictive count data distributions in retail sales forecasting. *International Journal of Forecasting*, 32(3):788–803, 2016.

[12] Nikolaos Kourentzes. Intermittent demand forecasts with neural networks. *International Journal of Production Economics*, 143(1):198–206, 2013.

[13] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The M4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1):54–74, 2020.

[14] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The M5 accuracy competition: Results, findings and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.

[15] Pablo Montero-Manso and Rob J. Hyndman. Principles and algorithms for forecasting groups of time series: Locality and globality. *International Journal of Forecasting*, 37(4):1632–1653, 2021.

[16] Fotios Petropoulos and Nikolaos Kourentzes. Forecast combinations for intermittent demand. *Journal of the Operational Research Society*, 66(6):914–924, 2015.

[17] David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3): 1181–1191, 2020.

[18] Gordon K. Smyth and Bent Jørgensen. Fitting Tweedie's compound Poisson model to insurance claims data: Dispersion modelling. *ASTIN Bulletin*, 32(1):143–157, 2002.

[19] Ralph D. Snyder, J. Keith Ord, and Adrian Beaumont. Forecasting the intermittent demand for slow-moving inventories: A modelling approach. *International Journal of Forecasting*, 28(2):485–496, 2012.

[20] Aris A. Syntetos and John E. Boylan. The accuracy of intermittent demand estimates. *International Journal of Forecasting*, 21(2):303–314, 2005.

[21] Aris A. Syntetos, John E. Boylan, and J. D. Croston. On the categorization of demand patterns. *Journal of the Operational Research Society*, 56(5):495–503, 2005.

[22] Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000.